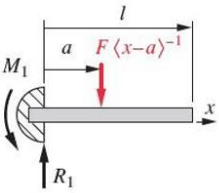
Beam Tables

(*a*) Cantilever beam with concentrated loading

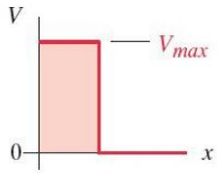


$$R_1 = F$$

$$M_1 = Fa$$

Loading

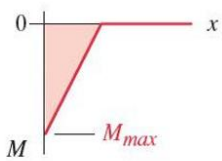
$$q = M_1\langle x\rangle^{-2} + R_1\langle x\rangle^{-1} - F\langle x-a\rangle^{-1}$$



$$V_{max} = R_1 = F$$

$$V = M_1\langle x\rangle^{-1} + R_1 - F\langle x-a\rangle^{0}$$
$$= F\left(1 - \langle x-a\rangle^{0}\right)$$

Shear

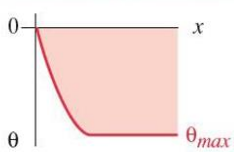


$$M_{max} = -Fa$$

$$\boxed{\text{when } a = l: \ M_{max} = -Fl}$$

$$M = -M_1 + R_1 x - F\langle x-a\rangle^{1}$$
$$= F\left(-a + x - \langle x-a\rangle^{1}\right)$$

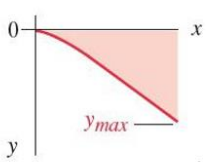
Moment



$$\theta = \frac{1}{EI}\begin{pmatrix} -M_1 x + \frac{R_1}{2}x^2 \\ -\frac{F}{2}\langle x-a\rangle^2 \end{pmatrix}$$

$$\theta = \frac{F}{2EI}\left(-2ax + x^2 - \langle x-a\rangle^2\right)$$

Slope



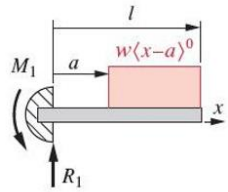
$$y_{max} = \frac{Fa^2}{6EI}(a - 3l)$$

$$\boxed{\text{when } a = l: \ y_{max} = -\frac{Fl^3}{3EI}}$$

$$y = \frac{1}{EI}\left(-\frac{M_1}{2}x^2 + \frac{R_1}{6}x^3 - \frac{F}{6}\langle x-a\rangle^3\right)$$
$$= \frac{F}{6EI}\left(x^3 - 3ax^2 - \langle x-a\rangle^3\right)$$

Deflection

(*b*) Cantilever beam with uniformly distributed loading

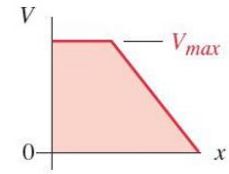


$$R_1 = w(l-a)$$

$$M_1 = \frac{w}{2}\left(l^2 - a^2\right)$$

Loading

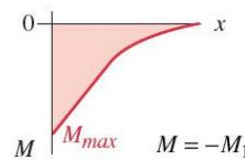
$$q = M_1\langle x\rangle^{-2} + R_1\langle x\rangle^{-1} - w\langle x-a\rangle^{0}$$



$$V_{max} = R_1 = w(l-a)$$

$$V = M_1\langle x\rangle^{-1} + R_1 - w\langle x-a\rangle^{1}$$
$$= w\left[(l-a) - \langle x-a\rangle^{1}\right]$$

Shear

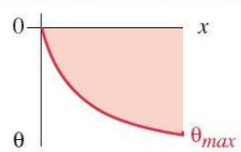


$$M_{max} = M_1 = \frac{w}{2}\left(l^2 - a^2\right)$$

$$\boxed{\text{when } a = 0: \ M_{max} = \frac{wl^2}{2}}$$

$$M = -M_1 + R_1 x - w\langle x-a\rangle^{2}$$
$$= \frac{w}{2}\left[2(l-a)x - \left(l^2 - a^2\right) - 2\langle x-a\rangle^2\right]$$

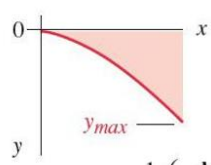
Moment



$$\theta = \frac{1}{EI}\begin{pmatrix} -M_1 x + \frac{R_1}{2}x^2 \\ -\frac{w}{6}\langle x-a\rangle^3 \end{pmatrix}$$

$$\theta = \frac{w}{6EI}\left(3(l-a)x^2 - 3\left(l^2 - a^2\right)x - \langle x-a\rangle^3\right)$$

Slope



$$y_{max} = \frac{w}{24EI}\left(-3l^4 + 4a^3 l - a^4\right)$$

$$\boxed{\text{when } a = 0: \ y_{max} = -\frac{wl^4}{8EI}}$$

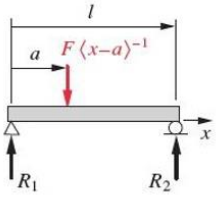
$$y = \frac{1}{EI}\left(-\frac{M_1}{2}x^2 + \frac{R_1}{6}x^3 - \frac{w}{24}\langle x-a\rangle^4\right)$$
$$= \frac{w}{24EI}\left(4(l-a)x^3 - 6\left(l^2 - a^2\right)x^2 - \langle x-a\rangle^4\right)$$

Deflection

FIGURE B-1

Cantilever Beams with Concentrated or Distributed Loading. Note: < > Denotes a Singularity Function

(*a*) Simply supported beam with concentrated loading

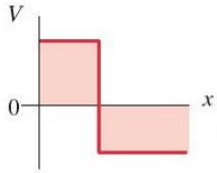


$$R_1 = F\left(1 - \frac{a}{l}\right)$$

$$R_2 = F\left(\frac{a}{l}\right)$$

Loading

$$q = R_1\langle x\rangle^{-1} - F\langle x-a\rangle^{-1} + R_2\langle x-l\rangle^{-1}$$

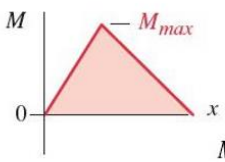


$$V_{max} = \mathrm{MAX}(R_1, R_2)$$

$$V = R_1 - F\langle x-a\rangle^0 + R_2\langle x-l\rangle^0$$

$$= F\left(1 - \frac{a}{l} - \langle x-a\rangle^0\right)$$

Shear



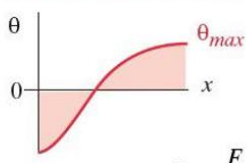
$$M_{max} = Fa\left(1 - \frac{a}{l}\right)$$

$$\boxed{\text{when } a = \frac{l}{2}: \quad M_{max} = \frac{Fl}{4}}$$

$$M = R_1 x - F\langle x-a\rangle^1 + R_2\langle x-l\rangle^1$$

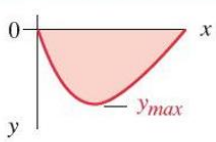
$$= F\left[\left(1 - \frac{a}{l}\right)x - \langle x-a\rangle^1\right]$$

Moment



$$\theta = \frac{F}{2EI}\left\{\begin{array}{l}\left(1 - \frac{a}{l}\right)x^2 - \langle x-a\rangle^2 \\ \quad + \frac{a}{3l}\left(-a^2 + 3al - 2l^2\right)\end{array}\right\}$$

Slope

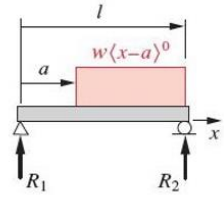


$$y_{max} = \frac{F}{3EI}\left(2a^3 - \frac{a^4}{l} - la^2\right)$$

$$y = \frac{F}{6EI}\left\{\begin{array}{l}\left(1 - \frac{a}{l}\right)x^3 - \langle x-a\rangle^3 \\ \quad + \frac{a}{l}\left(-a^2 + 3al - 2l^2\right)x\end{array}\right\}$$

Deflection

(*b*) Simply supported beam with uniformly distributed loading

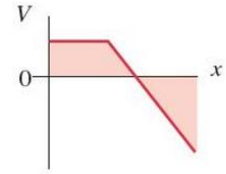


$$R_1 = \frac{w}{2l}(l-a)^2$$

$$R_2 = \frac{w}{2l}\left(l^2 - a^2\right)$$

Loading

$$q = R_1\langle x\rangle^{-1} - w\langle x-a\rangle^0 + R_2\langle x-l\rangle^{-1}$$

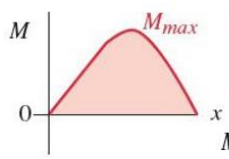


$$V_{max} = \mathrm{MAX}(R_1, R_2)$$

$$V = R_1 - w\langle x-a\rangle^1 + R_2\langle x-l\rangle^0$$

$$= w\left[\frac{1}{2l}(l-a)^2 - \langle x-a\rangle^1\right]$$

Shear

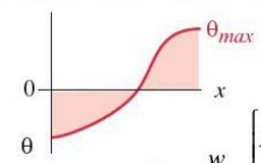


$$\boxed{\text{when } a = 0: \quad M = \frac{wx}{2}(l-x)}$$

$$M = R_1 x - \frac{w}{2}\langle x-a\rangle^2 + R_2\langle x-l\rangle^1$$

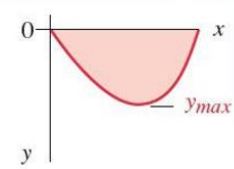
$$= \frac{w}{2}\left[\frac{x}{l}(l-a)^2 - \langle x-a\rangle^2\right]$$

Moment



$$\theta = \frac{w}{24EI}\left\{\begin{array}{l}\frac{6x^2}{l}(l-a)^2 - 4\langle x-a\rangle^3 \\ \quad + \frac{1}{l}\left[(l-a)^4 - 2l^2(l-a)^2\right]\end{array}\right\}$$

Slope



$$\boxed{\begin{array}{l}\text{when } a = 0: \\ \quad y = \frac{wx}{24EI}\left(2lx^2 - x^3 - l^3\right)\end{array}}$$

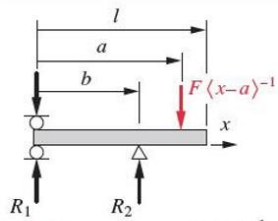
$$y = \frac{w}{24EI}\left\{\begin{array}{l}\frac{2x^3}{l}(l-a)^2 - \langle x-a\rangle^4 \\ \quad + \frac{x}{l}\left[(l-a)^4 - 2l^2(l-a)^2\right]\end{array}\right\}$$

Deflection

FIGURE B-2

Simply Supported Beams with Concentrated or Distributed Loading. Note: < > Denotes a Singularity Function

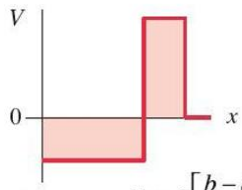
(*a*) Overhung beam with concentrated loading



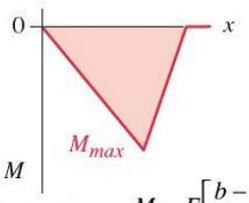
$$R_1 = F\left(\frac{b-a}{b}\right)$$

$$R_2 = F\left(\frac{a}{b}\right)$$

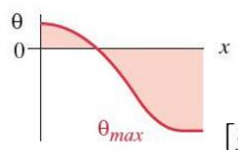
Loading $\quad q = R_1\langle x\rangle^{-1} - F\langle x-a\rangle^{-1} + R_2\langle x-b\rangle^{-1}$



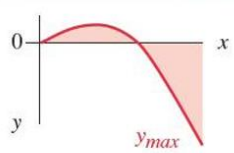
Shear $\quad V = F\left[\frac{b-a}{b}\langle x\rangle^0 + \frac{a}{b}\langle x-b\rangle^0 - \langle x-a\rangle^0\right]$



Moment $\quad M = F\left[\frac{b-a}{b}\langle x\rangle^1 + \frac{a}{b}\langle x-b\rangle^1 - \langle x-a\rangle^1\right]$

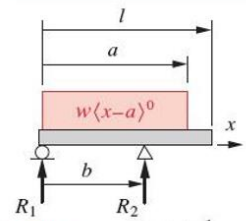


Slope $\quad \theta = \frac{F}{2EI}\left[\frac{b-a}{b}\langle x\rangle^2 + \frac{a}{b}\langle x-b\rangle^2 - \langle x-a\rangle^2 + \frac{b}{3}(a-b)\right]$



Deflection $\quad y = \frac{F}{6EI}\left[\frac{b-a}{b}\langle x\rangle^3 + \frac{a}{b}\langle x-b\rangle^3 - \langle x-a\rangle^3 + b(a-b)x\right]$

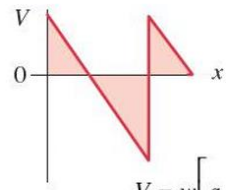
(*b*) Overhung beam with uniformly distributed loading



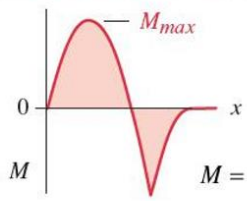
$$R_1 = wa\left(1 - \frac{a}{2b}\right)$$

$$R_2 = \frac{wa^2}{2b}$$

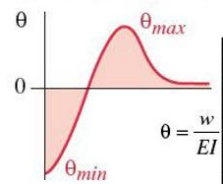
Loading $\quad q = R_1\langle x\rangle^{-1} - w\langle x\rangle^0 + w\langle x-a\rangle^0 + R_2\langle x-b\rangle^{-1}$



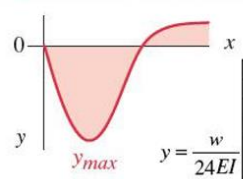
Shear $\quad V = w\left[a - x + \langle x-a\rangle^1 + \frac{a^2}{2b}\left(\langle x-b\rangle^0 - 1\right)\right]$



Moment $\quad M = \frac{w}{2}\left[\left(2a - \frac{a^2}{b}\right)x - x^2 + \langle x-a\rangle^2 + \frac{a^2}{b}\langle x-b\rangle^1\right]$



Slope $\quad \theta = \frac{w}{EI}\left[\frac{1}{24}\left(2a^2b - 4ab^2 + b^3 - \frac{1}{b}\langle b-a\rangle^4\right) + \left(\frac{a}{2} - \frac{a^2}{4b}\right)x^2 - \frac{1}{6}x^3 + \frac{a^2}{4b}\langle x-b\rangle^2 + \frac{1}{6}\langle x-a\rangle^3\right]$



Deflection $\quad y = \frac{w}{24EI}\left[\left(2a^2b - 4ab^2 + b^3 - \frac{1}{b}\langle b-a\rangle^4\right)x + \left(4a - \frac{2a^2}{b}\right)x^3 - x^4 + \langle x-a\rangle^4 + \frac{2a^2}{b}\langle x-b\rangle^3\right]$

FIGURE B-3

Overhung Beams with Concentrated or Distributed Loading. Note: < > Denotes a Singularity Function